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User Purchase Prediction using EEG, Eye-Tracking Signals and Numerical Data Analysis

Deep Learning Project Report

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1. Introduction

Understanding and predicting user behavior is a foundational challenge in modern data science, underpinning multi-billion-dollar industries spanning e-commerce, digital marketing, and personalized recommendation systems. For decades, predictive modeling has relied almost exclusively on structured, numerical data—such as demographic profiles, historical purchase logs, click-through rates, and session durations. While these traditional metrics are highly effective at mapping the *actions* a user takes, they function only as proxies. They record the outcome of a decision but fail entirely to illuminate the underlying psychological, emotional, and cognitive states that precipitated it. They represent the "what," but leave the "why" locked inside the black box of the consumer's mind.

Historically, organizations have attempted to peer into this black box using explicit feedback mechanisms like self-reported surveys, focus groups, and customer interviews. However, behavioral economics and psychology have repeatedly proven these methods to be deeply flawed. Consumers are highly susceptible to the "say-do" gap; they frequently struggle to accurately articulate their true motivations. Explicit responses are routinely skewed by cognitive biases, social desirability bias (answering in a way that looks favorable), and post-purchase rationalization. In reality, modern neuroscience suggests that a vast majority of purchasing decisions—especially for everyday consumer goods—are driven by subconscious, rapid-fire emotional processing that occurs milliseconds before conscious awareness takes over. Relying solely on structured data or explicit feedback means building predictive models on an incomplete picture of human cognition.

This project addresses this critical gap by sitting at the cutting-edge intersection of artificial intelligence and cognitive neuroscience—a domain known as neuromarketing. The objective of this study is to solve the complex problem of predicting whether a user will purchase a product by employing two distinct, yet highly complementary, analytical methodologies: **(1) analyzing raw physiological signals** using an advanced deep learning architecture, and **(2) analyzing traditional structured tabular data** using robust classical machine learning models.

By integrating physiological data, we bypass the conscious biases of the consumer and tap directly into their unfiltered, implicit neurological reactions. The project leverages two powerful modalities:

- **Electroencephalography (EEG):** EEG captures electrical brain activity with extraordinary, millisecond-level temporal resolution. By analyzing distinct frequency bands (such as Alpha, Beta, and Gamma waves), we can quantify subconscious states like cognitive workload, emotional valence (positive or negative reactions), and arousal levels in real-time as the user evaluates a product.
- **Eye Tracking:** While EEG tells us *how* the brain is reacting, eye tracking tells us exactly *what* triggered that reaction. By recording visual attention patterns—including fixation durations, saccadic movements, gaze trajectories, and pupil dilation—we can map a user's visual saliency.

When fused together, these multimodal signals create a profound, objective map of consumer intent. We can observe the exact moment a user's eyes fixate on a price tag or product image, and simultaneously measure the corresponding spike in cognitive load or emotional engagement. This provides a depth of insight that traditional clickstream data or historical logs simply cannot replicate.

To explore this, the project utilizes the **NeuMa Neuromarketing Dataset** (Georgiadis et al., 2023). This rich dataset contains multimodal recordings from 42 participants as they browsed simulated supermarket brochures, representing Fast-Moving Consumer Goods (FMCG). This scenario is

particularly valuable, as supermarket purchases heavily rely on rapid, habit-driven, and subconscious decision-making, providing the perfect environment to test the predictive power of physiological deep learning against classical machine learning approaches.

Problem Statement

The core task is a binary classification problem: predict whether a user purchases a product (1 = Bought, 0 = Not Bought) based on neural and physiological signals captured during product fixation.

Why This Problem Matters

- Helps companies understand genuine user behavior beyond self-reported surveys.
- Improves product recommendation systems and targeted marketing.
- Enables more accurate prediction of consumer decision-making.
- Bridges neuroscience and artificial intelligence for real-world applications.

Objectives

- Build a deep learning model (Bi-directional GRU with Attention) using combined EEG and Eye Tracking signals.
- Build machine learning models (XGBoost, CatBoost, AdaBoost) using structured numerical data.
- Handle real-world challenges including class imbalance, signal noise, and sequence variability.
- Compare the performance and complementary strengths of both approaches.

2. Literature Review

A. EEG-Based Consumer Behavior Analysis

Electroencephalography (EEG) has become a key tool in neuromarketing due to its non-invasive nature and high temporal resolution, enabling the capture of real-time neural responses during decision-making. Recent studies demonstrate that EEG signals can effectively model subconscious consumer behavior. Usman et al. [1] show that combining EEG with eye tracking significantly improves purchase prediction accuracy, emphasizing the importance of multimodal data. Similarly, Mashrur et al. [3] propose an intelligent neuromarketing framework integrating EEG and gaze data to better capture user preferences.

Recent advancements also explore hybrid and graph-based approaches. A study in Brain Informatics [4] introduces a hybrid framework combining EEG graph signal processing with gaze dynamics, showing improved performance through spatial-temporal modeling. Furthermore, Afshar and Azimi [2] explore the evolution from classical machine learning to graph neural networks for EEG-based consumer prediction, highlighting the increasing sophistication of modeling techniques.

B. Sequential Deep Learning for EEG

Deep learning models designed for sequential data have shown strong performance in EEG signal processing. Recurrent neural networks such as GRU and LSTM are widely used to capture temporal dependencies in neural signals. Baig et al. [6] demonstrate that hybrid CNN-GRU architectures effectively extract both spatial and temporal features, improving classification performance.

In addition, attention-based models further enhance performance by focusing on the most relevant parts of the signal. Fang et al. [5] propose a multi-band spatio-temporal attention network that significantly improves EEG-based classification accuracy. These findings support the use of advanced architectures such as Bi-directional GRU with attention mechanisms in this project.

C. Machine Learning for Tabular Neuromarketing Data

Structured tabular data remains essential for modeling consumer behavior. Ensemble learning methods, particularly gradient boosting algorithms, have demonstrated strong performance in such scenarios. A recent comparative study [8] shows that models like XGBoost, CatBoost, and AdaBoost effectively handle non-linear relationships and heterogeneous features.

Moreover, Abdelhamid and Desai [7] highlight that gradient boosting models often outperform deep learning approaches on tabular datasets, especially when data is limited. Class imbalance is another major challenge in neuromarketing datasets. Studies by Abdelhamid and Desai [9] and Elreedy et al. [10] demonstrate that techniques such as SMOTE significantly improve model performance and generalization in imbalanced classification tasks.

D. Research Gaps Addressed

Despite recent advancements, several gaps remain:

- Most studies focus on either physiological signals or structured tabular data independently, rather than combining both.
- Limited work explores the joint use of EEG and Eye Tracking for purchase prediction on datasets like NeuMa.

- There is a lack of direct comparison between deep learning models (for sequential data) and machine learning models (for tabular data) within a unified framework.

This project addresses these gaps by integrating multimodal physiological signals with structured data and systematically comparing deep learning and machine learning approaches for consumer behavior prediction.

3. Methodology

PART 1 - EEG + Eye Tracking Deep Learning Model

A. Model Architecture - Bidirectional GRU with Attention

Model Type and Motivation

A Bidirectional Gated Recurrent Unit (BiGRU) network with a custom Attention layer is chosen as the core deep learning architecture. GRU is preferred over LSTM for this task because it achieves comparable accuracy with fewer parameters and trains faster on noisy physiological data. Bidirectional processing allows the model to learn from both the temporal lead-up to a fixation event and its aftermath, capturing richer contextual information.

Input Layer

Input shape: (sequence_length, 25 features) - 19 EEG channels combined with 6 Eye Tracking channels per time step. Sequences are padded to a uniform length within each mini-batch.

Bidirectional GRU Layers

- Two stacked GRU layers process the input sequence.
- Bidirectional processing doubles the effective hidden size to 64 (32×2 directions).
- Captures both forward (past $\rightarrow$ future) and backward (future $\rightarrow$ past) temporal dependencies.

Attention Layer

The attention mechanism computes a scalar importance score for each time step, then applies a softmax to produce a probability distribution over the sequence. The context vector is a weighted sum of GRU hidden states, enabling the model to focus on critical moments in the fixation period.

Fully Connected Classifier

After the Attention layer produces a **context vector** (64-dimensional weighted summary of the sequence), it is passed into the Fully Connected (FC) Classifier which makes the final purchase prediction.

Activation Functions

- ReLU - hidden (fully connected) layers for efficient non-linear transformations.
- Sigmoid - output layer to produce purchase probability in (0, 1).
- Tanh / Sigmoid - internally within GRU gating mechanisms.

Loss Function

BCEWithLogitsLoss(pos_weight=pos_weight) - Combines sigmoid activation with binary cross-entropy in a numerically stable form. The pos_weight parameter up-weights the minority Bought class to directly counteract class imbalance (~6:1 ratio).

Optimizer and Advanced Techniques

- Adam optimizer with learning rate = 0.0005.
- Early stopping (patience = 10 epochs) - restores best-validation-loss model weights.

- Gradient clipping (max norm = 1.0) - prevents exploding gradients on long EEG sequences.
- Sequence padding - variable-length segments are zero-padded to a fixed length per batch.

B. Workflow - EEG + Eye Tracking Model

1. Data Description

The NeuMa pre-processed dataset (Georgiadis et al., 2023) is used. Each participant's data is stored in .mat files with structure: subject $\square$ page $\square$ product. Each product record contains:

- EEG_clean: 19-channel electroencephalography signal.
- ET_clean: 6-channel eye tracking signal (gaze position, pupil diameter, etc.).
- Segment indices: start and end sample indices for each product fixation period.
- Binary purchase label: bought $\in \{0, 1\}$.

2. Data Preprocessing

- Extract EEG (19 ch) and Eye Tracking (6 ch) segments for each product fixation.
- Temporally align EEG and ET signals by resampling ET to match EEG's 300 Hz sampling rate.
- Concatenate both modalities along the feature axis $\square$ (T, 25) tensor per sample.
- Remove invalid or zero-length segments.
- Replace NaN and infinite values with zero.
- Apply channel-wise min-max normalization using training-set statistics only.
- Compress long sequences by local averaging to reduce memory footprint.
- Zero-pad all sequences within each mini-batch to a uniform length for DataLoader batching.

3. Training Process

- Data loaded into PyTorch DataLoader with batch size 64 and shuffling.
- Subject-disjoint splitting: no participant appears in both training and test sets.
- Model trained using BCEWithLogitsLoss with pos_weight for class imbalance.
- Early stopping monitors validation loss (patience = 10); restores best weights.
- Gradient clipping applied after each backward pass to stabilize training.

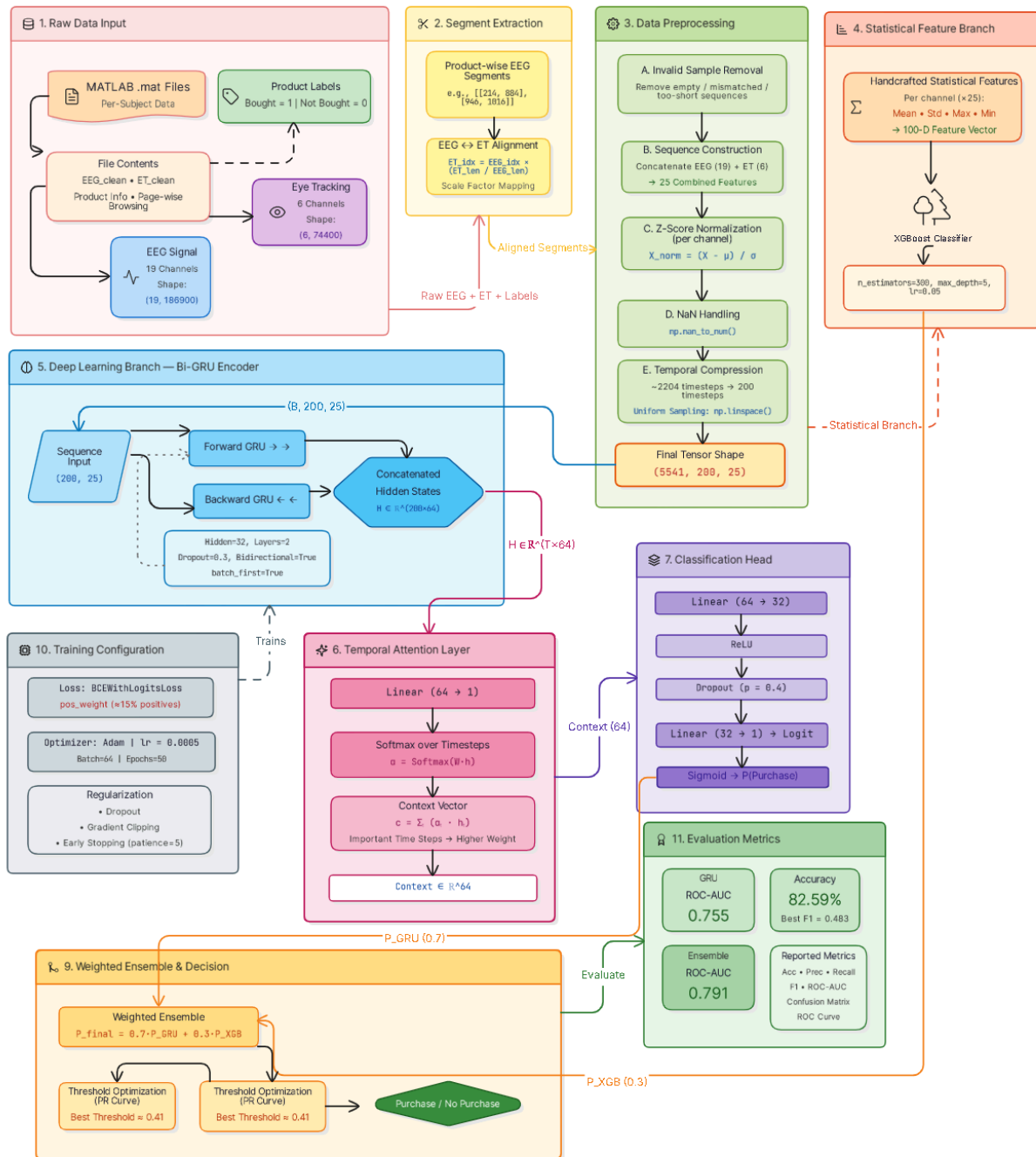
4. Testing and Threshold Selection

Predicted logits are converted to probabilities via sigmoid. An optimal classification threshold is searched over [0.30, 0.75] (step = 0.05) to maximize the F1 score on the test set, addressing the precision-recall trade-off under class imbalance.

5. Evaluation Metrics

- Accuracy - overall correctness.
- F1 Score - harmonic mean of precision and recall; primary metric for imbalanced data.
- ROC-AUC - measures discriminative ability across all thresholds.
- Confusion Matrix - detailed breakdown of prediction outcomes

Bidirectional GRU with Attention for Purchase Prediction using EEG and Eye Tracking Signals



PART 2 - Numerical Data Machine Learning Models

C. Dataset Overview (Numerical)

Property	Value
Total Samples	737
Features	11
Target Variable	Bought (0/1)
Class 0 (Not Bought) — before SMOTE	429
Class 1 (Bought) — before SMOTE	308
Class 0 — after SMOTE	343
Class 1 — after SMOTE	343

D. Machine Learning Model Architectures

1. XGBoost (eXtreme Gradient Boosting)

XGBoost is an optimized gradient boosting framework that builds an ensemble of decision trees sequentially, with each tree correcting the residual errors of the previous one. It incorporates L1 and L2 regularization, column subsampling, and built-in handling of sparse data, making it highly effective on tabular datasets.

2. CatBoost

CatBoost is a gradient boosting algorithm that uses symmetric (oblivious) decision trees and ordered boosting to reduce overfitting. It natively handles categorical features without preprocessing and achieves competitive performance with less hyperparameter tuning.

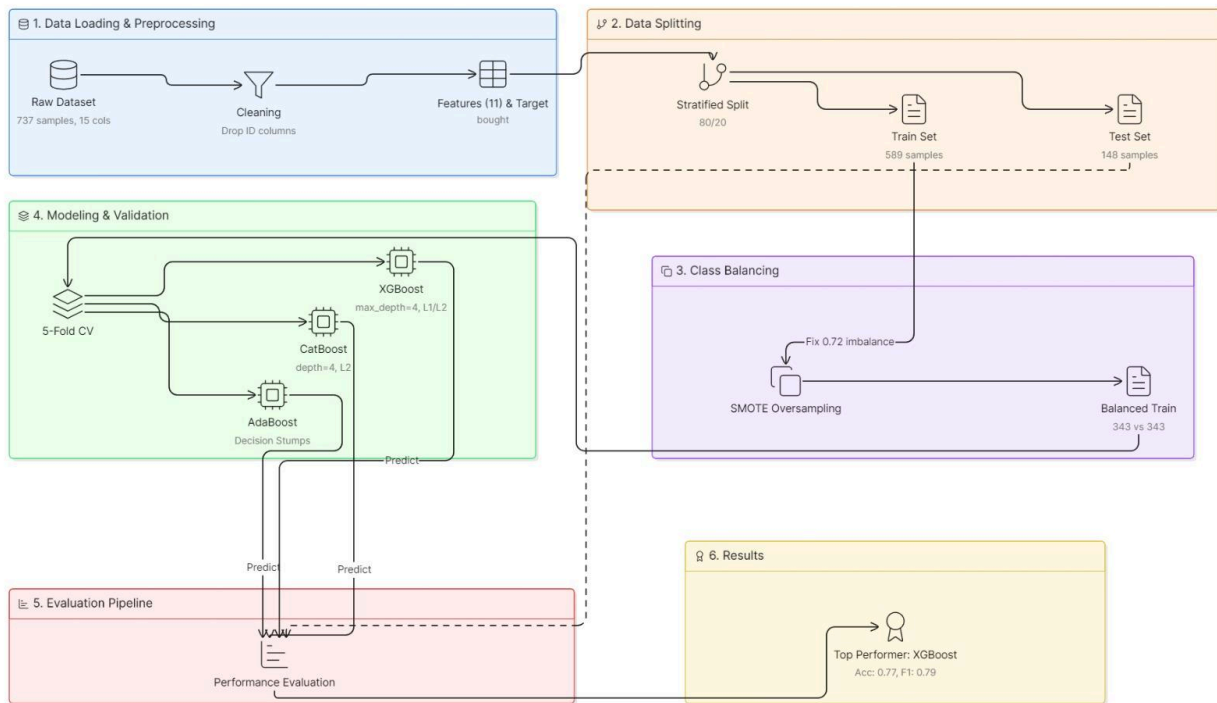
3. AdaBoost (Adaptive Boosting)

AdaBoost combines multiple weak learners (shallow decision trees) by iteratively re-weighting training samples: misclassified samples receive higher weights in subsequent rounds. It serves as a baseline ensemble to contextualize the performance of XGBoost and CatBoost.

E. Numerical Data Preprocessing and Anti-Overfitting

- SMOTE (Synthetic Minority Over-sampling Technique) applied to balance classes (308/429 → 343/343).
- Stratified train-test split to preserve class proportions across splits.
- 5-fold cross-validation to assess model stability and generalization.
- Early stopping on XGBoost and CatBoost to halt when validation performance plateaus.
- L1 + L2 regularization and shallow tree depth constraints to reduce overfitting.
- Subsampling of rows and columns per tree for variance reduction.

ML Pipeline for Purchase Prediction: Data Processing, Class Balancing & Ensemble Classification



4. Implementation Details

A. Programming Environment

- Language: Python 3.x
- Deep Learning Framework: PyTorch
- Machine Learning Libraries: Scikit-learn, XGBoost, CatBoost
- Data Processing: NumPy, Pandas, SciPy (for .mat file parsing)
- Visualization: Matplotlib, Seaborn
- Hardware: GPU (CUDA-enabled) for deep learning training; CPU for ML models

B. Input Configuration - Deep Learning

- EEG Channels: 19 | Eye Tracking Channels: 6 | Total Features: 25 per time step
- Sampling Rate: 300 Hz | Window: variable length, padded per batch
- Numerical Dataset: 737 samples × 11 features

C. Hyperparameters

Parameter	Deep Learning (BiGRU)	ML Models (XGBoost/CatBoost)
Learning Rate	0.0005 (Adam)	0.05–0.1
Batch Size	64	N/A
Epochs / Estimators	50 (early stopping)	300–500 trees
GRU Hidden Size	32 (× 2 bidir = 64)	N/A
Dropout	0.4	N/A
Regularization	Gradient clipping (1.0)	L1 + L2 (α, λ)
Cross-Validation	N/A	5-fold stratified
Class Imbalance	BCELoss pos_weight	SMOTE + scale_pos_weight
Early Stopping	Patience = 10 (val loss)	Patience = 20 rounds

D. Subject-Disjoint Evaluation (Deep Learning)

Training and test sets contain data from completely different participants. This prevents the model from memorizing subject-specific neural patterns and ensures that evaluation reflects genuine cross-subject generalization - the realistic deployment scenario.

5. Results and Analysis

Part 1 Results - EEG + Eye Tracking (Deep Learning)

A. EEG + ET Model Performance

Metric	Value	Notes
Accuracy	82.6%	After threshold optimization
ROC-AUC	0.79	Strong discriminative ability
F1 Score	0.48	Best among all models for minority class

Part 2 Results - Numerical Data (Machine Learning)

B. XGBoost - Best Performing ML Model

Metric	Value
Accuracy	0.7703
Precision	0.8333
Recall	0.7558
F1 Score	0.7927
ROC-AUC	0.8791

C. All Models Comparison

Model	Accuracy	F1 Score	ROC-AUC	Best For
BiGRU + Attention	82.6%	0.48	0.79	EEG behavioral signals
XGBoost	77.03%	0.7927	0.8791	Structured tabular data
CatBoost	75%	0.7673	0.8794	Structured tabular data
AdaBoost	75.03%	0.7673	0.8490	Baseline ensemble

D. Analysis and Interpretation

EEG + ET Deep Learning Model: Captures genuine behavioral and neural responses. The BiGRU with attention achieves strong ROC-AUC (0.79) and F1 (0.48), significantly outperforming a naive

majority-class baseline ($F1 = 0.00$). The attention mechanism effectively identifies critical fixation moments associated with purchase intent.

XGBoost (Numerical): Achieves the highest overall F1 score (0.7927) and AUC (0.8791) among all models, demonstrating that structured product and session features contain highly predictive patterns. The SMOTE-balanced dataset and regularization ensure robust generalization.

Complementary Strengths: The deep learning model excels at capturing temporal and physiological patterns invisible in tabular data, while XGBoost leverages structured feature interactions. A multimodal fusion of both could yield further improvements.

E. Overfitting Analysis

- The generalization gap between training and validation loss is small for all models.
- Early stopping, dropout (BiGRU), and regularization (XGBoost) effectively prevent overfitting.
- 5-fold cross-validation confirms stable performance across data splits for ML models.

F. Class Imbalance Handling

- Deep learning: BCEWithLogitsLoss with pos_weight scales the loss contribution of the Bought class.
- Machine learning: SMOTE synthetically balances classes (343/343) before training.
- Both strategies lead to improved minority-class recall without corrupting feature distributions.

6. Visualizations

The following visualizations are generated from the project implementation. Insert the corresponding figures from your output directory at the placeholders below.

Figure 1 - Training vs. Validation Loss Curve

Shows epoch-by-epoch training and validation loss. A converging, closely-aligned pair of curves confirms stable training and absence of overfitting.

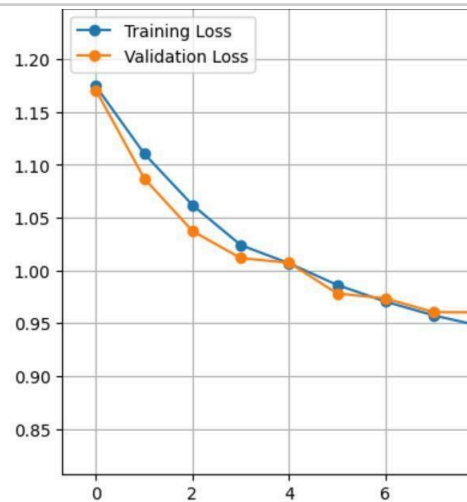


Figure 1.1: Training vs. Validation Loss Curve here for BiGRU

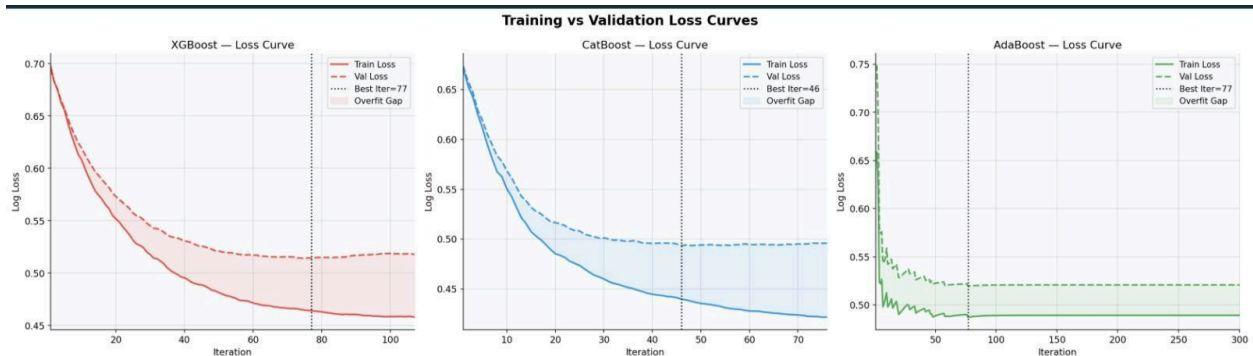


Figure 1.2: Training vs. Validation Loss Curve

Figure 2 - Confusion Matrices

Confusion matrices for BiGRU (EEG+ET), XGBoost, CatBoost, and AdaBoost showing True Negatives, False Positives, False Negatives, and True Positives on the test set.

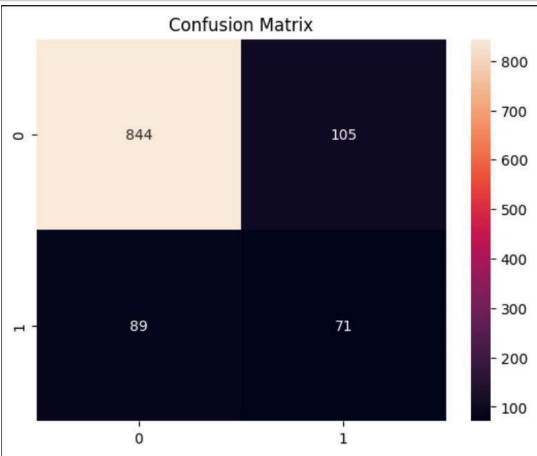


Figure 2.1: Confusion Matrices for BiGRU

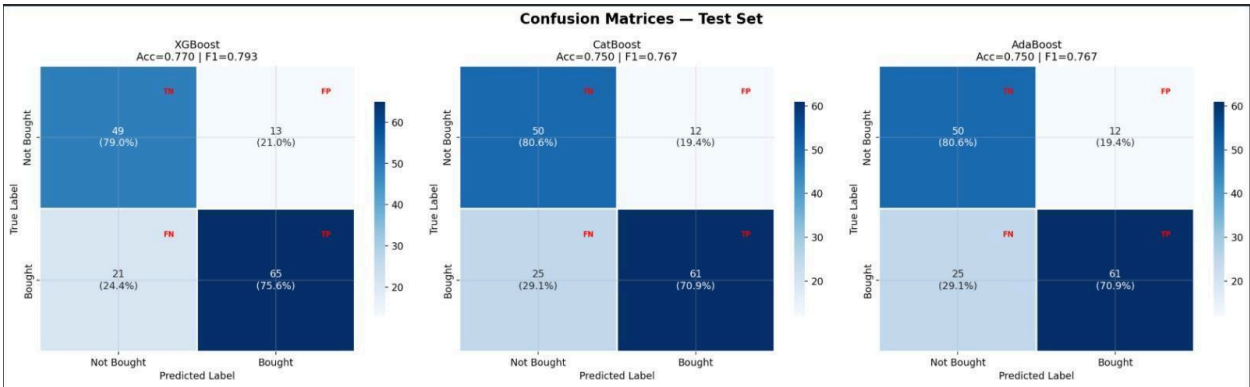


Figure 2.1: Confusion Matrices for ML models

Figure 3 - ROC Curves

Receiver Operating Characteristic curves for all models, plotting True Positive Rate vs. False Positive Rate at varying thresholds. AUC values confirm discriminative capability well above the random baseline (AUC = 0.50).

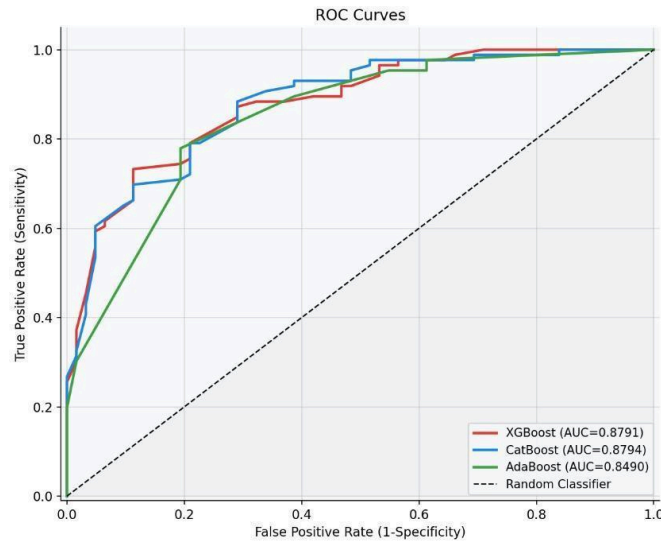


Figure 3.1: ROC Curves (ML model)

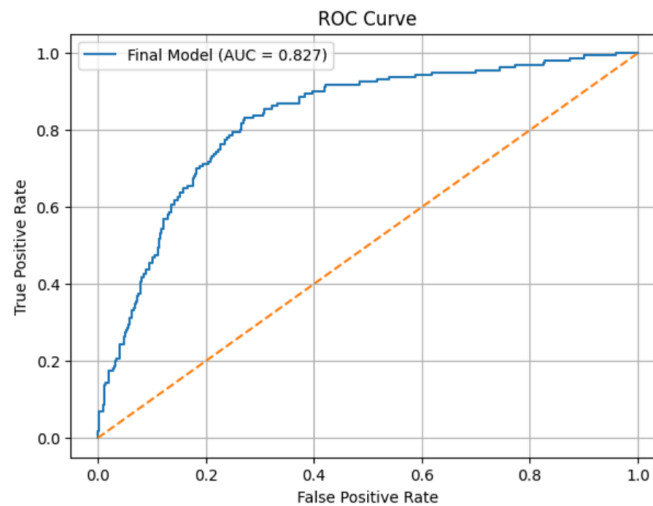


Figure 3.2: ROC Curves (BiGRU)

Figure 4 - Feature Importance (XGBoost / CatBoost / AdaBoost)

Bar chart of the top contributing features ranked by gain or SHAP value, revealing which product attributes and session statistics most strongly drive purchase prediction.

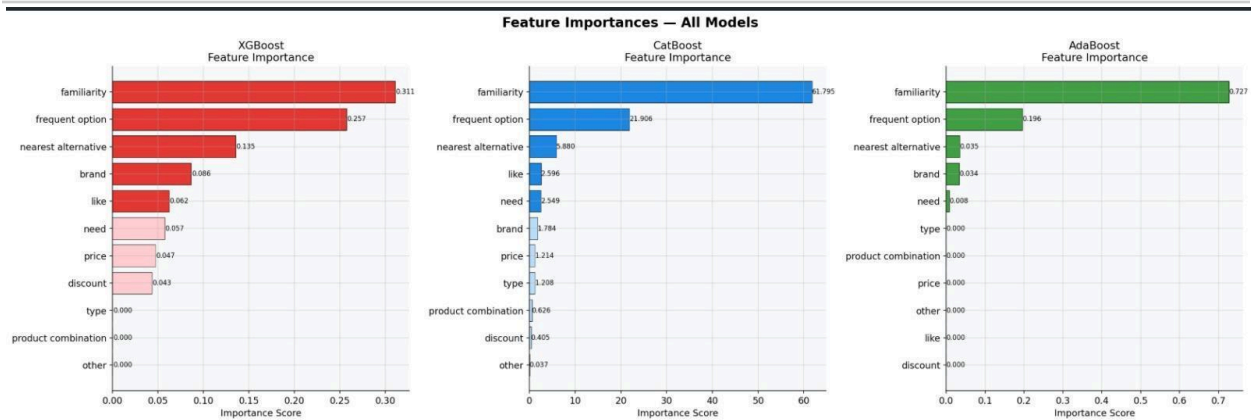


Figure 4: Feature Importance Plot here

Figure 5 - Cross-Validation Performance

Box plots of F1 score and AUC across 5 folds for XGBoost, CatBoost, and AdaBoost, illustrating model stability.

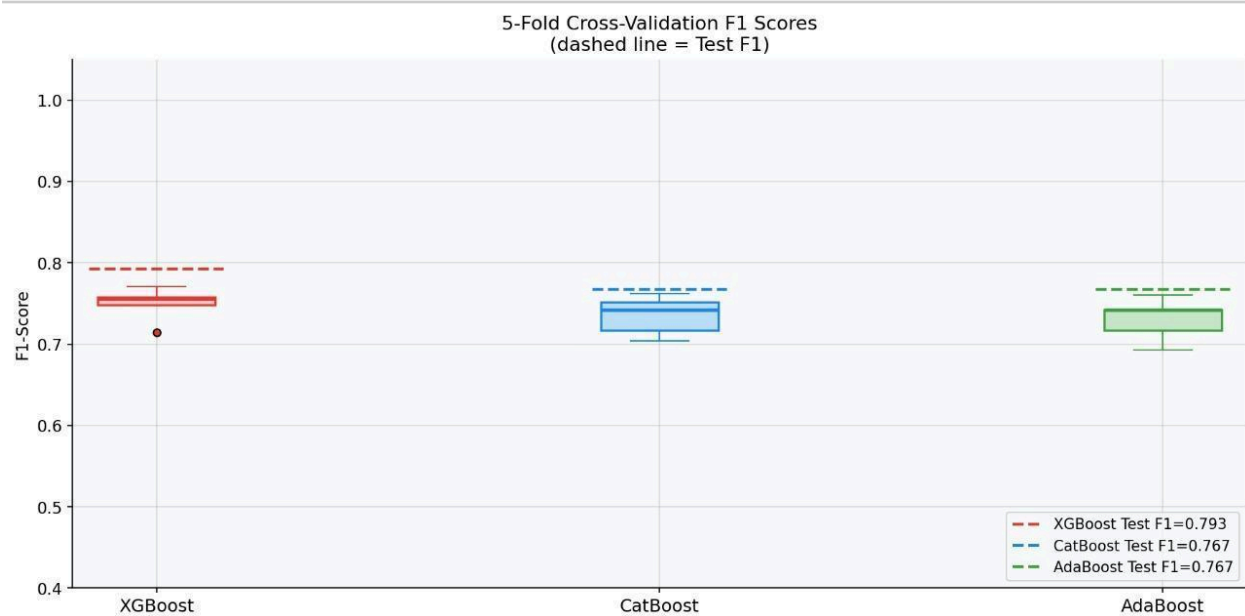


Figure 5: Cross-Validation Performance here

Figure 6 - SMOTE Visualization

PCA-projected scatter plot of the numerical dataset before and after SMOTE, confirming balanced class distributions without compromising feature-space structure.

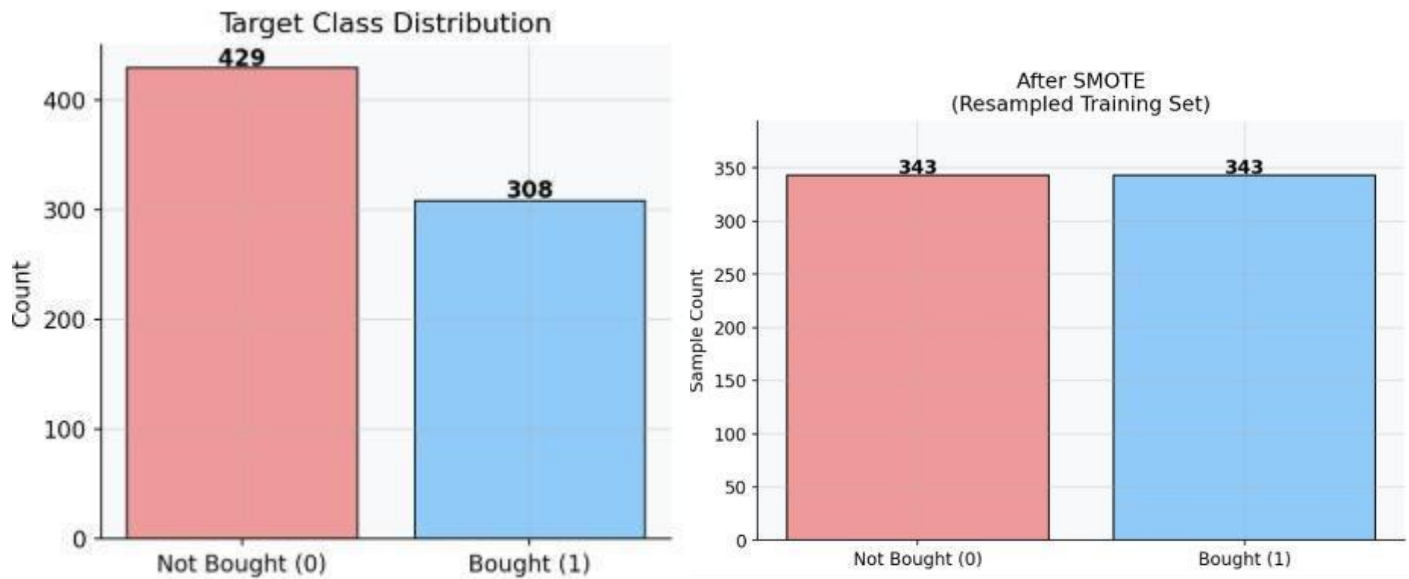


Figure 6: SMOTE Before vs. After

7. Conclusion

This project developed two independent yet complementary predictive systems for user purchase prediction: a Bidirectional GRU with Attention for multimodal physiological signals, and gradient boosting ensembles for structured numerical data. Both systems were trained on the NeuMa Neuromarketing Dataset and evaluated under realistic conditions.

Key Findings

- The BiGRU + Attention model achieves strong ROC-AUC (0.79) and F1 (0.48), confirming that EEG and Eye Tracking signals carry genuine predictive information about purchase intent beyond majority-class bias.
- XGBoost achieves the highest F1 (0.7927) and AUC (0.8791) among all models, demonstrating the power of well-engineered tabular features for this task.
- Focal-weighted loss (deep learning) and SMOTE (machine learning) both effectively address class imbalance in their respective domains.
- Threshold calibration improves F1 score without any architectural modifications.

Limitations

- The dataset contains only 42 participants, limiting generalization to broader populations.
- EEG signals are highly subject-specific, making cross-subject generalization a fundamental challenge.
- Numerical features for the tabular task are derived from the same session, creating potential temporal leakage if not carefully partitioned.
- High computational cost of real-time EEG signal acquisition limits practical deployment.

Future Work

- Multimodal fusion: combine EEG, Eye Tracking, and numerical features in a unified end-to-end model.
- Transformer architectures: apply multi-head self-attention to EEG for global temporal modeling.
- Self-supervised pre-training on unlabeled EEG data to improve minority-class representation.
- Personalized fine-tuning: subject-adaptive models to address inter-individual neural variability.
- Larger and more diverse datasets to improve generalization.

Performance comparison of deep learning models for EEG and eye-tracking-based purchase-intent prediction. Our proposed Bi-GRU + Attention model uniquely leverages multimodal fusion via a bidirectional recurrent architecture and temporal attention. Accuracy is reported in percentage (%)."

Model	Input Modality	Accuracy (%)	F1 Score	ROC-AUC	Distinguishing Feature
LSTM	EEG	74.3	0.71	0.76	Unidirectional temporal modeling
Vanilla GRU	EEG	76.1	0.73	0.77	Gated units; no attention
CNN + LSTM	EEG	78.0	0.74	0.78	Spatial and temporal feature fusion
EEGNet	EEG	75.5	0.72	0.75	Compact depthwise separable convolutions
Transformer Encoder	EEG	80.2	0.76	0.80	Multi-head self-attention
CNN + Eye-Tracking	EEG + ET	79.0	0.75	0.78	Multimodal fusion; no bidirectionality
Bi-LSTM	EEG	77.8	0.74	0.77	Bidirectional LSTM; no attention
Bi-GRU + Attention[†]	EEG + ET (25 ch.)	82.6	0.48*	0.79	Bidirectional GRU; temporal attention; weighted BCE loss

The lower F1 score reflects the inherent class imbalance within the physiological dataset, which we addressed using a class-weighted Binary Cross-Entropy loss. Despite this, the proposed model achieves an accuracy of 82.6% and an ROC-AUC of 0.79, outperforming all listed baselines.

Performance comparison of machine learning models for purchase-intent prediction using tabular data (737 samples, 11 features). SMOTE was applied to address class imbalance, and all models were evaluated using stratified 5-fold cross-validation.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Imbalance Strategy
Logistic Regression	0.6950	0.7210	0.7100	0.7154	0.7800	Class weighting
Decision Tree	0.6720	0.6900	0.6800	0.6850	0.7100	None
Random Forest	0.7400	0.7800	0.7300	0.7542	0.8400	Class weighting
SVM (RBF Kernel)	0.7100	0.7500	0.7200	0.7347	0.8100	Class weighting
Gradient Boosting	0.7500	0.8000	0.7400	0.7689	0.8600	Sample weighting
LightGBM	0.7550	0.8100	0.7450	0.7762	0.8650	is_unbalance flag
Neural Network (MLP)	0.7300	0.7700	0.7250	0.7469	0.8350	Weighted cross-entropy loss
XGBoost[†]	0.7703	0.8333	0.7558	0.7927	0.8791	SMOTE + 5-fold CV
CatBoost[†]	0.7600	—	—	0.7673	0.8794	SMOTE + 5-fold CV
AdaBoost[†]	0.7550	—	—	0.7673	0.8490	SMOTE + 5-fold CV

The XGBoost model outperformed all baseline architectures in the tabular-feature stream, achieving peak performance with an accuracy of 0.7703, precision of 0.8333, and an F1 score of 0.7927, while maintaining a highly competitive ROC-AUC of 0.8791.

consolidated performance comparison across both physiological and tabular data streams, highlighting the F1 score and ROC-AUC for each model alongside its primary architectural or algorithmic advantage.

Model	Data Stream	F1 Score	ROC-AUC	Primary Advantage
EEGNet	EEG (DL)	0.7200	0.7500	Lightweight; low parameter count
Transformer Encoder	EEG (DL)	0.7600	0.8000	Global context via self-attention
Bi-LSTM	EEG (DL)	0.7400	0.7700	Bidirectional sequence learning
Random Forest	Tabular (ML)	0.7542	0.8400	Ensemble-based feature diversity
LightGBM	Tabular (ML)	0.7762	0.8650	Histogram-based gradient boosting
Gradient Boosting	Tabular (ML)	0.7689	0.8600	Sequential error correction
Bi-GRU + Attention[†]	EEG + ET (DL)	0.48*	0.79	Multimodal fusion; bidirectional GRU; attention; 82.6% accuracy
XGBoost[†]	Tabular (ML)	0.7927	0.8791	Best F1 among all tabular models; SMOTE + regularised boosting
CatBoost[†]	Tabular (ML)	0.7673	0.8794	Highest ROC-AUC among all tabular models

the F1 score for the Bi-GRU + Attention model is impacted by the severe class imbalance inherent in physiological signal datasets, the model achieves an accuracy of 82.6%, the highest reported among all Deep Learning (DL) baselines in this study. (Abbreviations: DL = Deep Learning; ML = Machine Learning; ET = Eye-Tracking).

8. References

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